Md. Nazmus Shakib

Ph.D. Student, Computer Engineering
University of Alabama in Huntsville

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Research Interests

Multimodal Data Fusion; Machine Intelligence; Cyber-Physical Systems; Autonomous Vehicles.

Education

Ph.D. in Computer Engineering — 2025 – Present
University of Alabama in Huntsville, USA — CGPA: 3.778/4.00

B.Sc. in Computer Science & Engineering — 2017 – 2022
Khulna University of Engineering & Technology (KUET), Bangladesh — CGPA: 3.75/4.00

Experience

Graduate Teaching Assistant — 2025 – Present
Dept. of Electrical & Computer Engineering, University of Alabama in Huntsville — Huntsville, AL

Graduate Student Researcher — Summer 2026
AARC Lab, University of Alabama in Huntsville — Huntsville, AL

Lecturer — 2022 – 2024
Dept. of Computer Science & Engineering, Green University of Bangladesh — Dhaka, Bangladesh

Publications

[1] **M. N. Shakib** and R. Bhadani, "Degradation-aware lightweight multimodal fusion for resilient headway estimation," in *27th International Conference on Cyber-Physical Systems (ICCPS)*, [Submitted], 2026.

[2] **M. N. Shakib** and R. Bhadani, "Towards a unified perspective on multimodal fusion in intelligent transportation systems," *Information Fusion*, 2026, [Submitted].

[3] **M. N. Shakib**, M. S. Uddin, and R. Bhadani, "String stability analysis of Followerstopper for mixed-autonomy vehicle platoons," in *2026 IEEE Intelligent Vehicles Symposium (IV)*, 2026.

[4] **M. N. Shakib**, M. A. H. Ifty, M. S. Uddin, and R. Bhadani, "Efficient CPU burst forecasting with low-latency machine learning models," in *IEEE SoutheastCon 2026*, 2026.

[5] M. R. Alom, M. A. Rahaman, M. P. Hossain, C. Mondal, **M. N. Shakib**, M. A. Habib, M. K. Hasan, and A. Ali, "Towards trustworthy brain stroke diagnosis using a lightweight explainable deep learning framework for CT imaging," *Scientific Reports*, 2026.

[6] M. S. Uddin, **M. N. Shakib**, and R. Bhadani, "Modeling electric vehicle car-following behavior: Classical vs machine learning approach," in *Proceedings of the IEEE Intelligent Transportation Systems Conference (ITSC)*, 2025.

[7] **M. N. Shakib**, M. Shamim, M. N. H. Shawon, M. K. F. Isha, M. Hashem, and M. Kamal, "An adaptive system for detecting driving abnormality of individual drivers using Gaussian mixture model," in *2021 5th International Conference on Electrical Engineering and Information Communication Technology (ICEEICT)*, IEEE, 2021, pp. 1–6.

[8] M. K. F. Isha, M. N. H. Shawon, M. Shamim, **M. N. Shakib**, M. Hashem, and M. A. S. Kamal, "A DNN based driving scheme for anticipatory car following using road-speed profile," in *2021 IEEE Intelligent Vehicles Symposium (IV)*, IEEE, 2021, pp. 496–501.

Selected Projects

CARLA and ROS 2 Multimodal Simulation Pipeline *Python, ROS 2, CARLA*
Developed a multimodal simulation pipeline bridging ROS 2 Humble and CARLA 0.9.16 to extract and synchronize camera–LiDAR data.

Platoon String Stability Framework *SUMO, Python*
Designed a SUMO simulation framework for mixed-autonomy traffic, featuring platoon string-stability metrics and wave-dampening controllers.

Parallel Bee Colony Optimization (BCO) for PID Tuning *C++, OpenMP*
Implemented an OpenMP-parallelized C++ Bee Colony Optimization meta-heuristic for PID tuning, analyzing execution speedup and scaling behavior against a MATLAB baseline.

Cruise Control System with LQR and PID *MATLAB, Simulink*
Modeled a cruise control system using PID and LQR controllers in MATLAB/Simulink.

Technical Skills

Programming	Python, C/C++, Java, MATLAB, Verilog
Frameworks & Systems	ROS 2, CARLA, SUMO, PyTorch, Simulink
Parallel & HPC	OpenMP, MPI, Alabama Supercomputing Cluster
Architecture & VLSI	Intel PIN, gem5, Design Compiler, IC Compiler, PrimeTime
Tools	Git, LaTeX, Quartus Prime, ModelSim

Awards & Achievements

Travel Grant, IEEE Intelligent Vehicles Symposium (IV) Jun 2026
Awarded to attend and present research at IEEE IV 2026.

Best Paper Award, 5th Int. Conf. on Electrical Engineering and Information & Communication Technology (ICEEICT 2021) Jan 2022
First prize in the Best Paper category among 103 accepted papers.

Dean's Award, Faculty of Electrical and Electronic Engineering, KUET 2018 – 2020
Received in the 2nd, 3rd and 4th years of undergraduate study.

Professional Service

Peer Reviewer, IEEE ITSC, IEEE CDC, and IEEE SoutheastCon 2025 – Present

Student Member, IEEE; IEEE Intelligent Transportation Systems Society (ITSS) 2024 – Present

Course Coordinator, Object-Oriented Programming and Data Mining, GUB 2023 – 2024

Selected Certifications

Certificate in Teaching and Learning (CTL), CETL, Green University of Bangladesh Oct 2022

Learning How to Learn, Coursera Oct 2020

IBM Data Science Professional Certificate, Coursera Jun 2020

Deep Learning Specialization, Coursera May 2020

Full list of certificates